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xApp Development using xDevSM

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Outline

1. O-RAN Overview
2. O-RAN E2 Interface
3. Open-Source Frameworks
4. xDevSM
5. xApp Deployment

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O-RAN Overview

O-RAN Introduction

O-RAN introduces a paradigm shift in the design, deployment, and operation of next-generation cellular networks by promoting openness, disaggregation, and programmability.

Designed to:

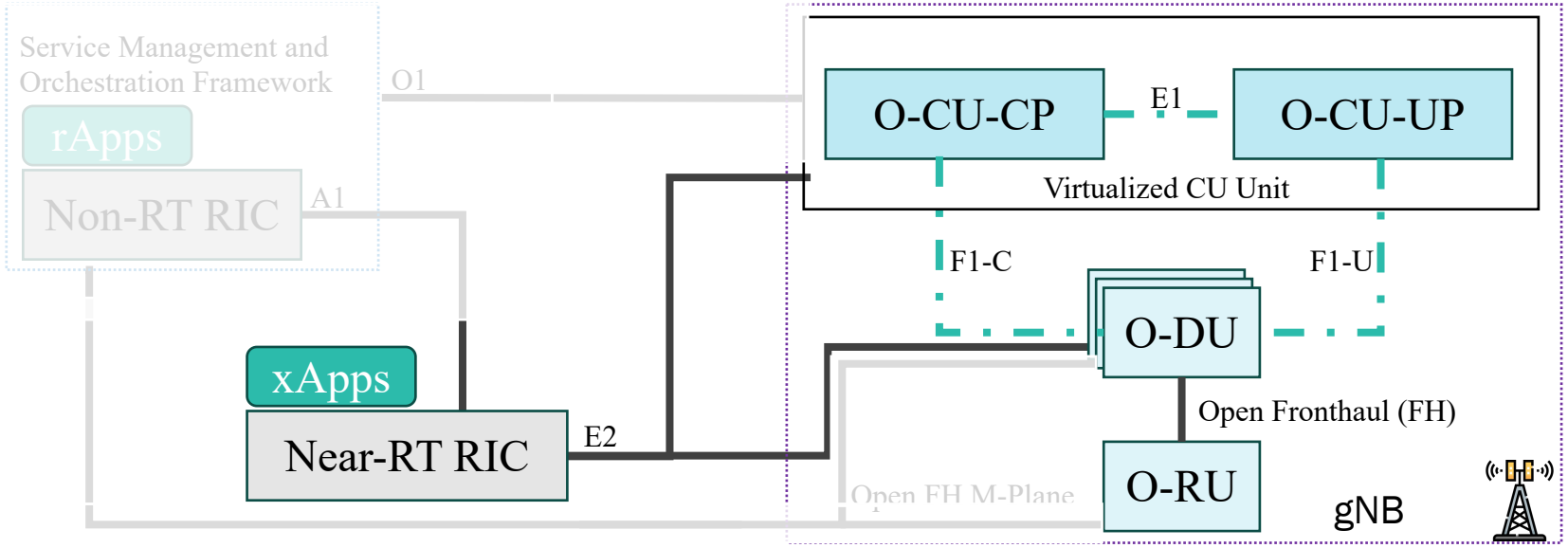
- Mitigate vendor lock-in through open interfaces
- Enable data-driven and ML-based optimization of the RAN
- Promote cloud-native and virtualized network architectures
- Support interoperable multi-vendor deployments

O-RAN Key Architectural Principles

Standardize an **architecture** and a **set of interfaces** to realize an Open-RAN.

Key principles:

- Disaggregation
- Radio Intelligent Controllers (RIC)
- Virtualization
- Open Interfaces



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O-RAN Working Groups (WGs)

Multiple Working Groups define **specifications** for O-RAN **architecture**, **interfaces**, and **cross-cutting functions.**

O-RAN ALLIANCE Specifications

Please note that as of 1st July 2022 for all versions of published O-RAN specifications the earlier IPR Policy will apply, except that the previous requirement for Adopters (as defined in the earlier IPR Policy) to agree to an O-RAN Adopter License Agreement to access and use Final Specifications shall no longer apply or be required for these Final Specifications after 1st July 2022.

Select specifications for download

Title/Designator	Type of Document	Release	Feature Package
architecture		-all-	-all-

Version	Work group	Published from - to	
-all-	-all-	From	To

Show latest

APPLY FILTER CLEAR FILTER

https://www.o-ran.org/specifications

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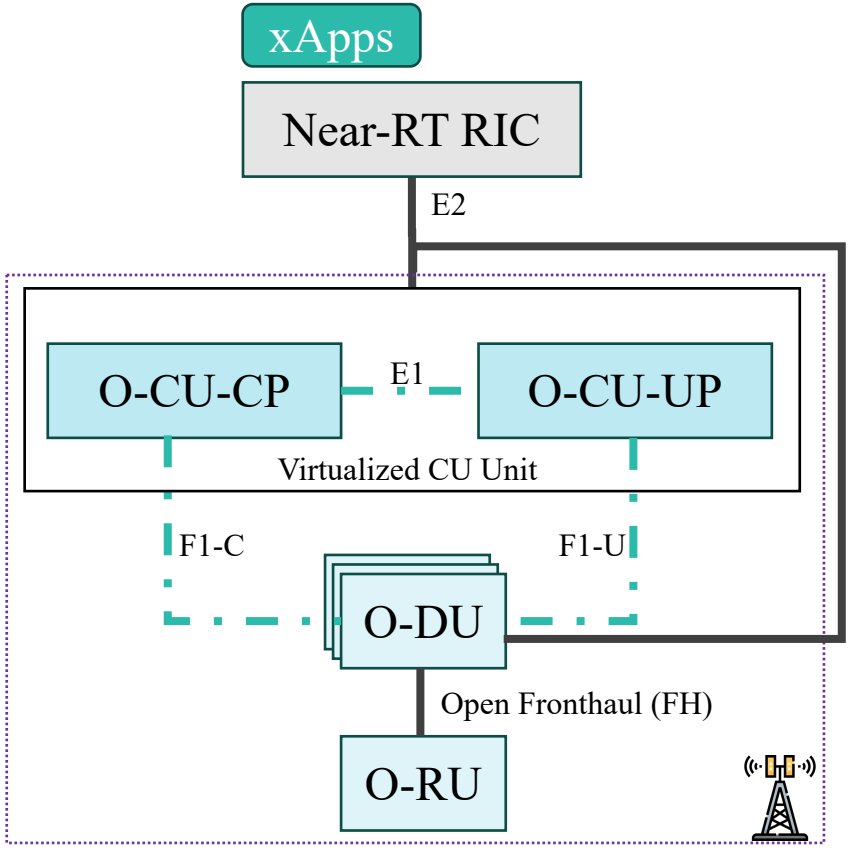
O-RAN E2 Interface

Near-RT RIC: E2 Interface Functional Role

- **Open interface** connecting the Near-RT RIC to E2 nodes (O-DU, O-CU-CP, O-CU-UP)
- **Near-real-time control timescale:** 10 ms – 1 s closed-loop RAN control
- **xApp capabilities via E2:** subscribe to RAN telemetry, receive event notifications, send control commands
- **Two-layer protocol design:**
  - E2AP handles signaling and connection lifecycle;
  - E2SM carries application-specific RAN content
- **E2 Setup procedure:** E2 node advertises supported RAN functions and corresponding E2SMs to the Near-RT RIC



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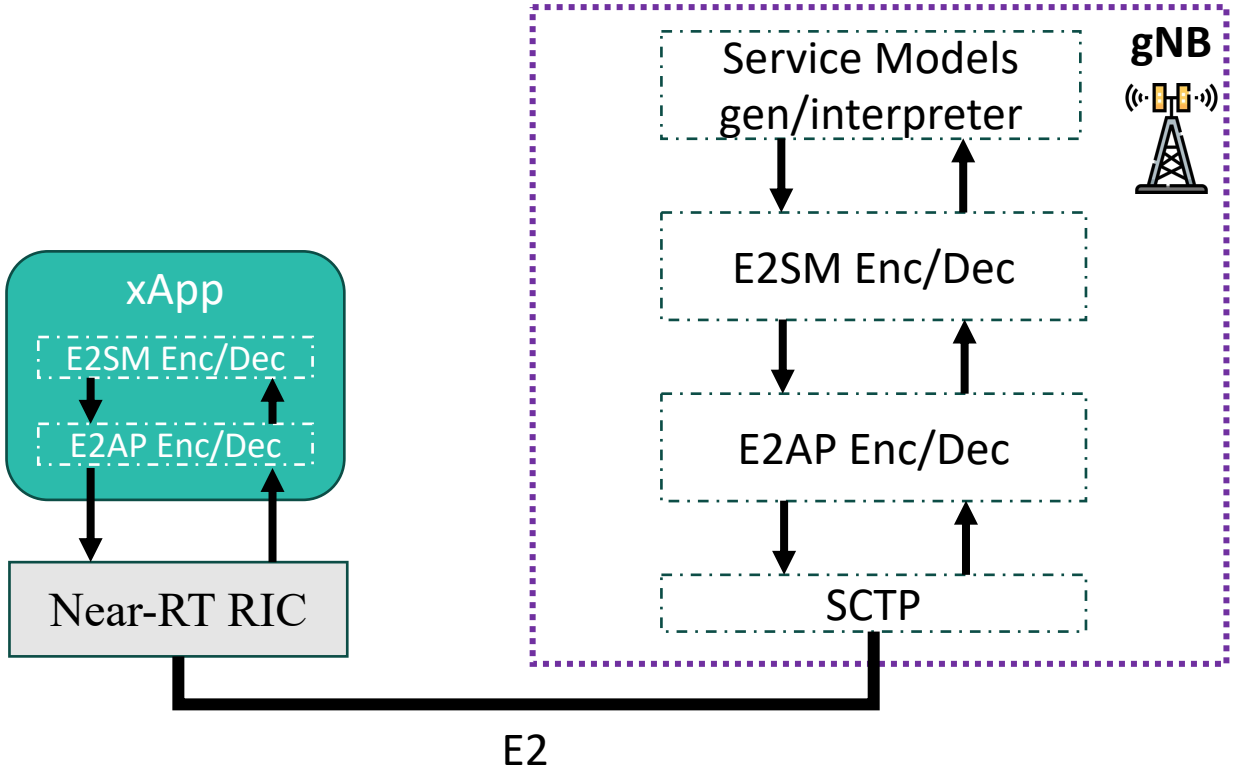
E2 Protocol Stack

Organized in two protocols:

- **E2 Application Protocol:** coordinates communications between Near-RT RIC and gNB
- **E2 Service Models:** Embedded within E2AP messages, E2SMs implement specific functionalities to support various applications and use cases



ASN.1 APER encoding: E2AP and E2SM messages are serialized using ASN.1 Packed Encoding Rules (PER); E2SM payloads are embedded as OCTET STRINGs inside E2AP messages

E2 Application Protocol (E2AP)

The **E2 Application Protocol** (E2AP) enables **standardized communications** between RIC Applications (xApps) and RAN functionalities exposed by E2 nodes

Provides:

- Signaling mechanism for subscription, reporting and control
- A defined set of RIC Service types
- A structured set of procedures for service management and message exchange

Initiated by	Elementary Procedure	Initiating Message	Successful Outcome	Unsuccessful Outcome
			Response message	Response message
Near-RT RIC	RIC Subscription	RIC SUBSCRIPTION REQUEST	RIC SUBSCRIPTION RESPONSE	RIC SUBSCRIPTION FAILURE
Near-RT RIC	RIC Subscription Delete	RIC SUBSCRIPTION DELETE REQUEST	RIC SUBSCRIPTION DELETE RESPONSE	RIC SUBSCRIPTION DELETE FAILURE
Near-RT RIC	RIC Subscription Modification	RIC SUBSCRIPTION MODIFICATION REQUEST	RIC SUBSCRIPTION MODIFICATION RESPONSE	RIC SUBSCRIPTION MODIFICATION FAILURE
E2 Node	RIC Subscription Modification Required	RIC SUBSCRIPTION MODIFICATION REQUIRED	RIC SUBSCRIPTION MODIFICATION CONFIRM	RIC SUBSCRIPTION MODIFICATION REFUSE
Near-RT RIC	RIC Subscription State Control	RIC SUBSCRIPTION STATE CONTROL REQUEST	RIC SUBSCRIPTION STATE CONTROL RESPONSE	RIC SUBSCRIPTION STATE CONTROL FAILURE
Near-RT RIC	RIC Subscription Audit	RIC SUBSCRIPTION AUDIT REQUEST	RIC SUBSCRIPTION AUDIT RESPONSE	RIC SUBSCRIPTION AUDIT FAILURE
E2 Node	RIC Assistance	RIC ASSISTANCE REQUEST	RIC ASSISTANCE RESPONSE	RIC ASSISTANCE FAILURE

E2 RIC Services

The E2 interface enables service-based interactions between E2 Nodes and the Near-RT RIC through a set of standardized RIC service types:

- **REPORT** – Subscription-based delivery of measurements and events from E2 Nodes to the Near-RT RIC
- **INSERT** – Event-triggered information reporting initiated by the E2 Node
- **CONTROL** – Near-Real-time control actions issued by the Near-RT RIC toward the E2 Node
- **POLICY** – Policy provisioning and enforcement guidance
- **QUERY** – On-demand information retrieval (RAN/UE-Related) from the E2 Node

Additional Service:

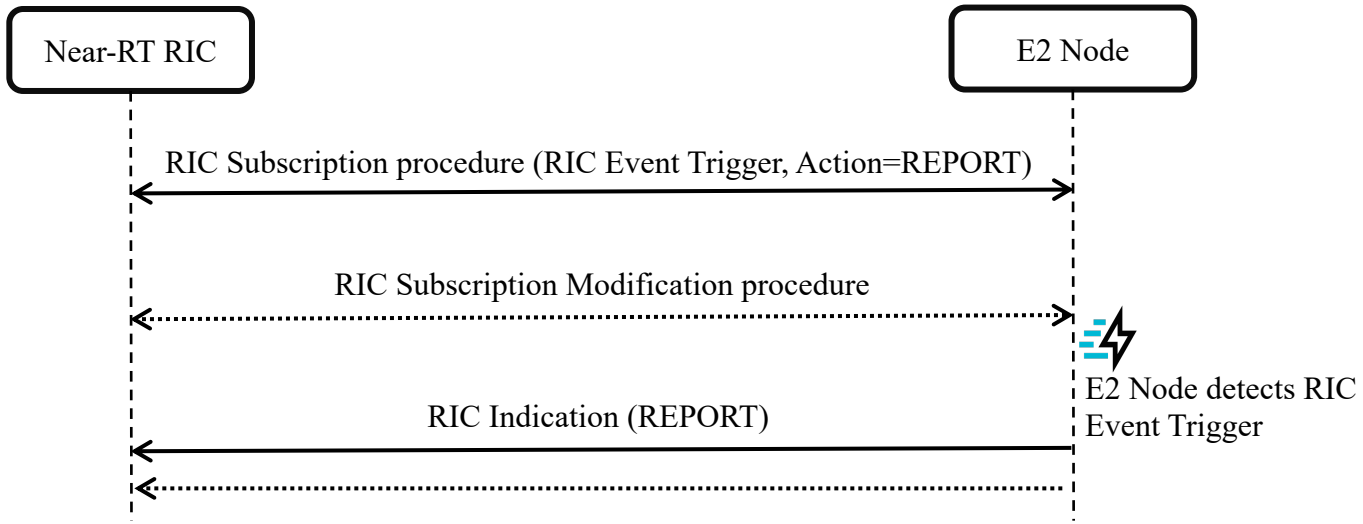
- **ASSISTANCE** – Enables E2 Nodes to request and leverage services provided by the Near-RT RIC

O-RAN.WG3.TS.E2GAP-R004-v07.00 - E2 General Aspects and Principles (E2GAP)

E2 RIC Services: REPORT

The Near-RT RIC initiates a **RIC Subscription Procedure** toward the E2 Node, specifying the measurements, events, or conditions of interest.

- Subscriptions can be updated through a Modification Procedure
- Associate procedure continues in the E2 Node, including any subsequent RIC actions



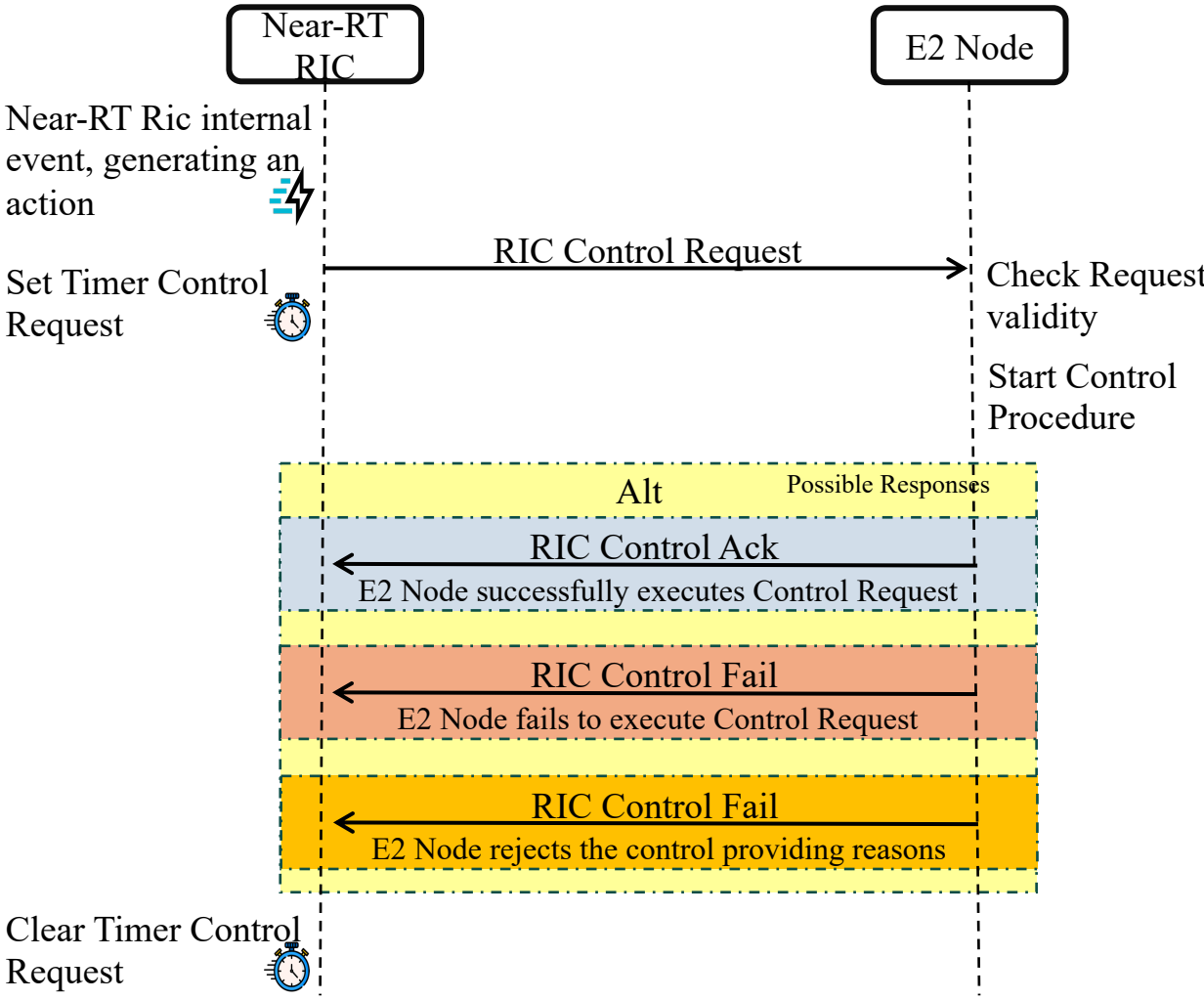
E2 RIC Service: CONTROL

Upon detection of an event (via an Event Trigger or Indication Message), the Near-RT RIC issues a **RIC Control Request** to the E2 Node

- The request specifies the control action and associated execution parameters

- If acknowledgment is requested, the E2 Node replies with a **RIC Control Acknowledge** message upon successful execution

- The E2 Node sends a **RIC Control Failure** message if the control action:
  - Cannot be executed (failure), or
  - Is rejected (e.g., due to policy or capability constraints)



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E2 Service Models (E2SM)

An O-RAN compliant Service Model defines different **styles**, for one or more RIC Services.

Four standard-compliant Service Models:

1. Key Performance Measurement (KPM)
   - expose and transmit a set of measurements
2. RAN Control (RC)
   - Radio Resource Management
3. Cell Configuration Control (CCC)
   - dynamic RAN parameter and tuning
4. Lower Layers Control (LLC)
   - PHY-layer report and optimization


```
{"ranName": "gnb_001_001_00000e00",
"connectionStatus": "DISCONNECTED",
"globalNbId": {
            "plmnId": "00F110",
            "nbId": "000000000000000000000111000000000"
 },
"nodeType": "GNB",
"gnb": {
"ranFunctions": [
            {
              "ranFunctionId": 2,
              "ranFunctionDefinition": "60304F52414E2…",
              "ranFunctionRevision": 3,
              "ranFunctionOid": "1.3.6.1.4.1.53148.1.3.2.2"
            },
            {
              "ranFunctionId": 3,
              "ranFunctionDefinition”: "6805804F52414E…",
              "ranFunctionRevision": 1,
              "ranFunctionOid": "1.3.6.1.4.1.53148.1.1.2.3"
            },
}
```


Each Service Model is identified at the Near-RT RIC by a **RAN Function ID.**

E2SM - KPM

This SM enables an E2 node to **periodically expose and transmit a set of measurements** to the Near-RT RIC.

- It supports only **REPORT RIC Service**
- Five styles defined for this service:
  1. E2 Node Measurement
  2. E2 Node Measurement for a single UE
  3. Condition-based, UE-level E2 Node Measurement
  4. **Common Condition-based, UE-level Measurement**
  5. E2 Node Measurements for multiple UE

O-RAN.WG3.E2SM-KPM-R003-v03.00 Technical Specification

E2SM - RC

- Essential for radio resource management (RRM) xApps
- Supports all RIC service types

Focus on the CONTROL RIC Service

- The CONTROL service supports **8 Service Styles**
- Each Service Style includes a set of associated control actions

Example:

- **Service Style 2 – Radio Resource Allocation Control**
  - Supports control actions related to the E2 node resource configuration and allocation

Service Style 2 – Radio Resource Allocation Control

Control Action ID	Control Action Name	Control Action description	Associated RAN Parameters
1	DRX parameter configuration	To control the configuration of DRX parameters	8.4.3.1
2	SR periodicity configuration	To control the configuration of SR periodicity parameters	8.4.3.2
3	SPS parameters configuration	To control the configuration of SPS parameters	8.4.3.3
4	Configured grant control	To control the configuration of uplink grants to the UE	8.4.3.4
5	CSI reports configuration	To control the configuration of CSI reports.	8.4.3.5
6	Slice-level PRB quota	To control the radio resource management policy for slice-specific PRB quota allocation	8.4.3.6
7	DMRS-Downlink configuration	To control the configuaration of DMRS for PDSCH	8.4.3.7
8	SRS Configuration	To control the configuration of the SRS transmissions	8.4.3.8
9	CSI Resource Configuration	To control the configuration of the CSI resources	8.4.3.9

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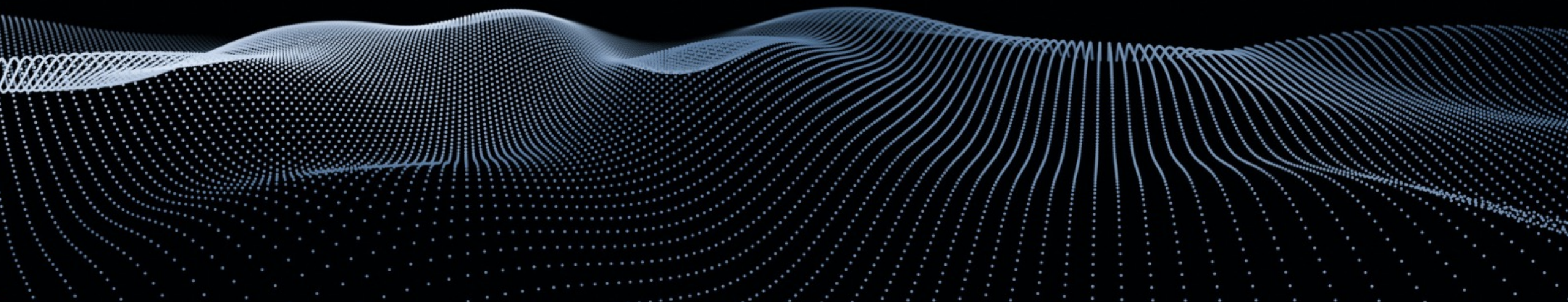
Open-Source Frameworks



Open-Source RAN Software

1. Openairinterface5G (next DURANTA)
   - Provides RAN (gNB), Core Network, and UE implementations
   - Widely used for research, prototyping, and experimental validation
   - Provides an implementation of the E2 Interface (RAN Side)
   - Supports O-RAN KPM Service Model

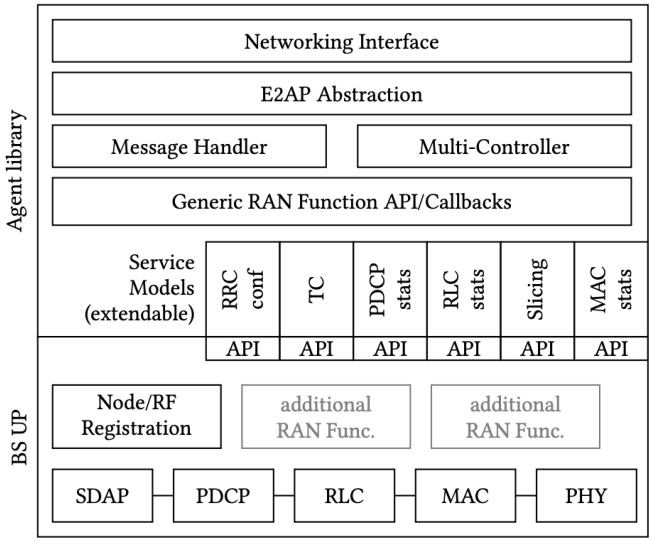
2. OCUDU (old srsRan)
   - RAN (gNB)
   - Widely used for research, prototyping, and experimental validation
   - Provides an implementation of the E2 Interface (RAN Side)
   - Supports O-RAN KPM and part of RC Service Model

Duranta: https://lfnetworking.org/projects/duranta/
OCUDU: https://ocudu.org/

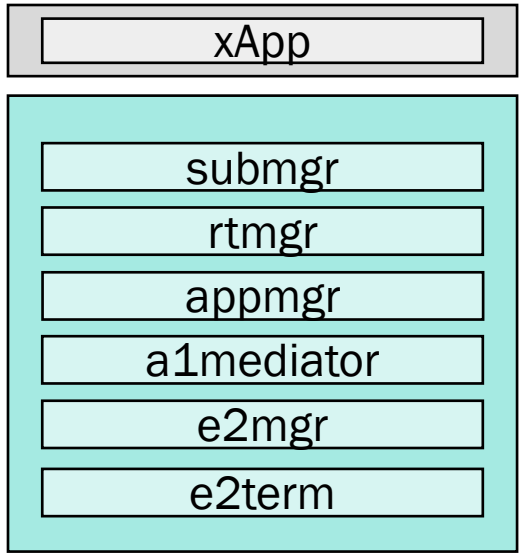
Near-RT RIC

Several development efforts are in progress to create a scaled-down versions of the Near-RT RIC:

- FlexRIC (monolithic)
- O-RAN Software Community (OSC) Near Real-Time RIC (microservices)
- Others…

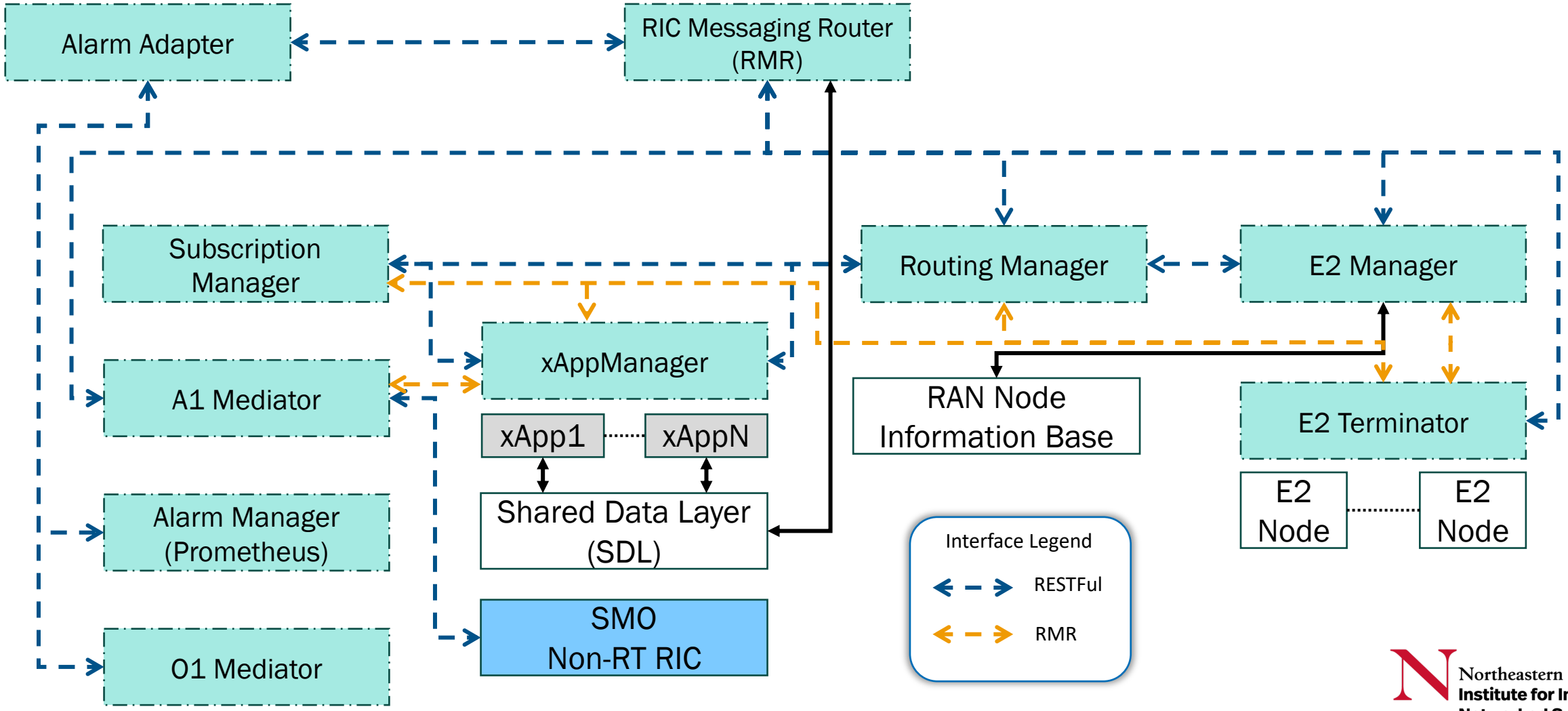


FlexRIC



OSC Near-RT RIC

O-RAN Software Community: Near-RT RIC



source: Near-RT RIC Comparison

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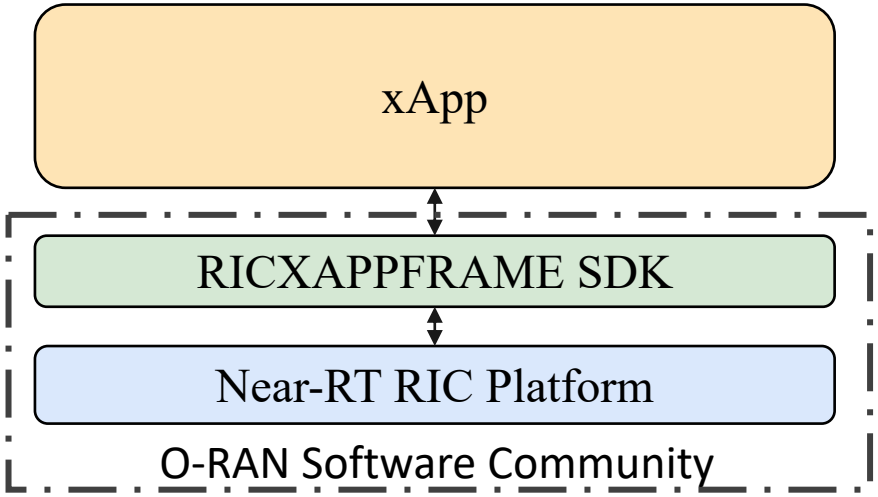
O-RAN Software Community: xApp Framework

The O-RAN Software Community (OSC) provides an xApp development framework that:

- Encodes Service Model data
- Enables communication with the OSC Near–Real-Time RIC via RMR APIs

However, xApp developers are still required to:

- Implement the xApp Logic
- Understand and navigate the structure of the required service models
- Implement E2SM Encoding/Decoding



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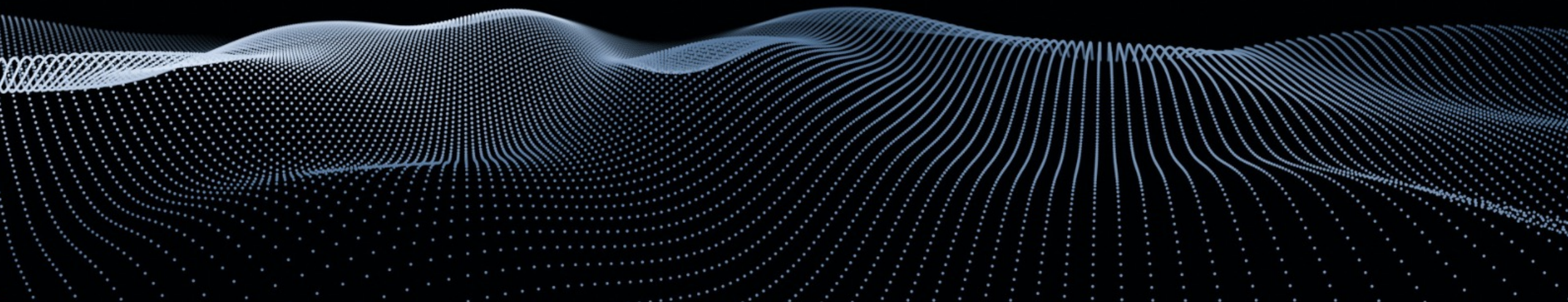
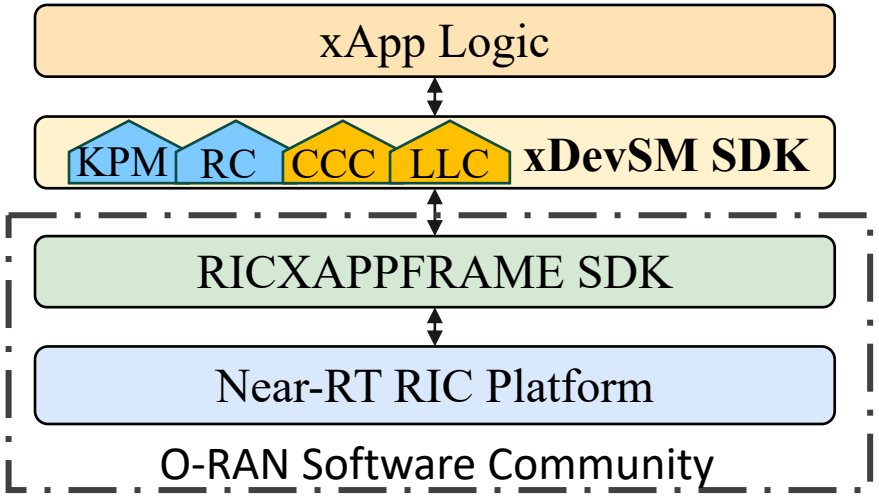
xDevSM Overview



xDevSM: A Flexible SM Framework for xApp Development

xDevSM a flexible and open-source framework to **streamline xApps development** for the OSC Near-RT RIC.

- Built on the OSC **Python** xApp framework
- Simple APIs to interact with E2 Service Models
- Support for different RAN implementations



xApp developers are still required to:

- Implement the xApp Logic
- ~~Understand and navigate the structure of the required service models~~
- ~~Implement E2SM Encoding/Decoding~~

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Supported E2SMs in RAN Software

RAN Stack	KPM (style 4)	RC Control Service			RC Report Service	
		RBC	RRAC	CMMC	UE Info	Msg Copy
OAI-2025	Y	Y (ID 1)P	-	-	Y	Y
OAI custom	Y	Y (ID 1)P	Y (ID 6)	Y (ID 3)	Y	-
srsRAN-2025 (now OCUDU)	Y	-	Y (ID 6)	Y (ID 3)	-	-

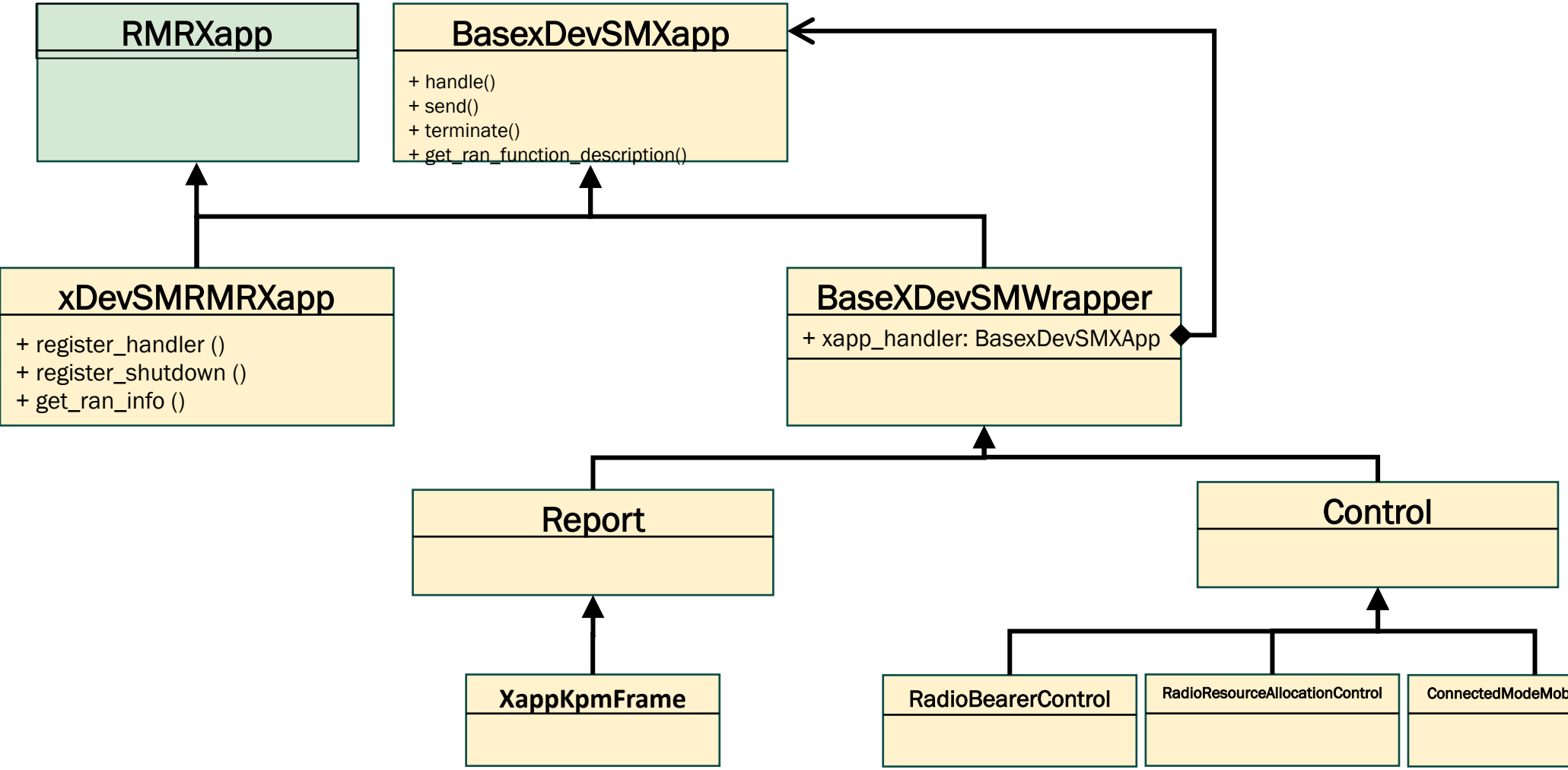
RBC: Radio Bearer Control
RRAC: Radio Resource Allocation Control
CMMC: Connected Mode Mobility Control

Supported E2SMs in xDevSM

Service Model	Service Type	Supported Actions
KPM (Key Performance Measurement)	Report	▪ Common Condition-based Measurement, UE-level Measurement
RC (RAN Control)	Control	▪ QoS Flow Mapping Configuration (RBC) ▪ Slice-level PRB Quota Action (RRAC) ▪ Handover Control (CMMC)

xDevSM Architecture Diagram

xDevSM follows the decorator design pattern



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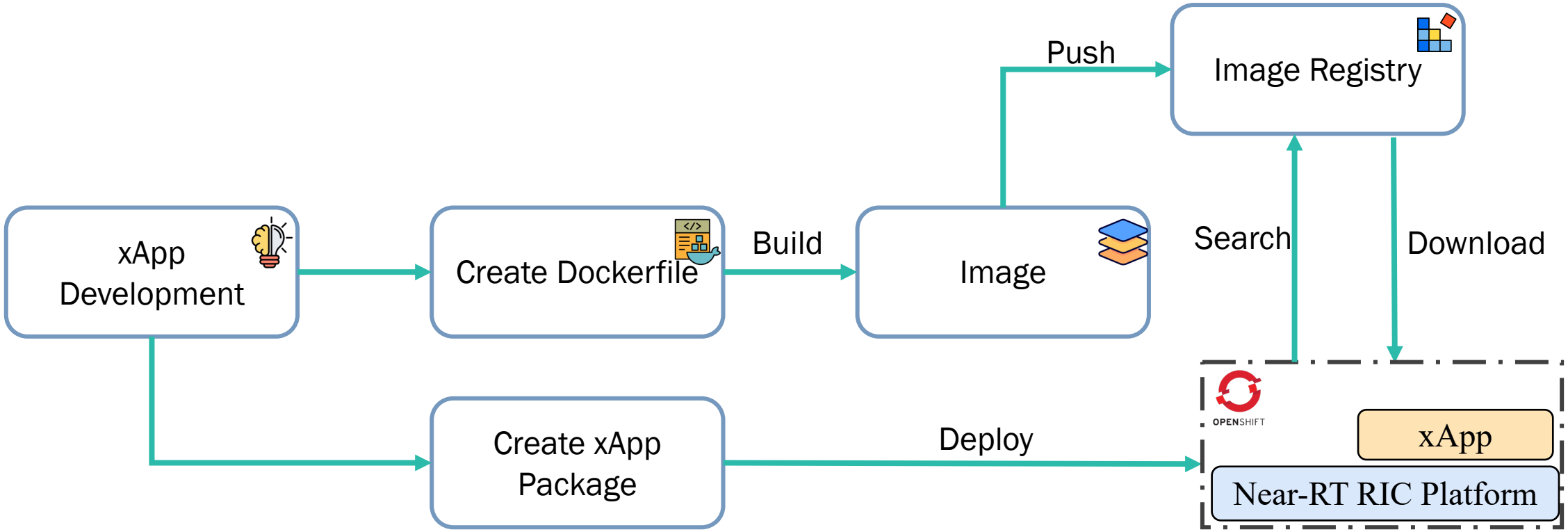
End-to-End xApp Development Workflow



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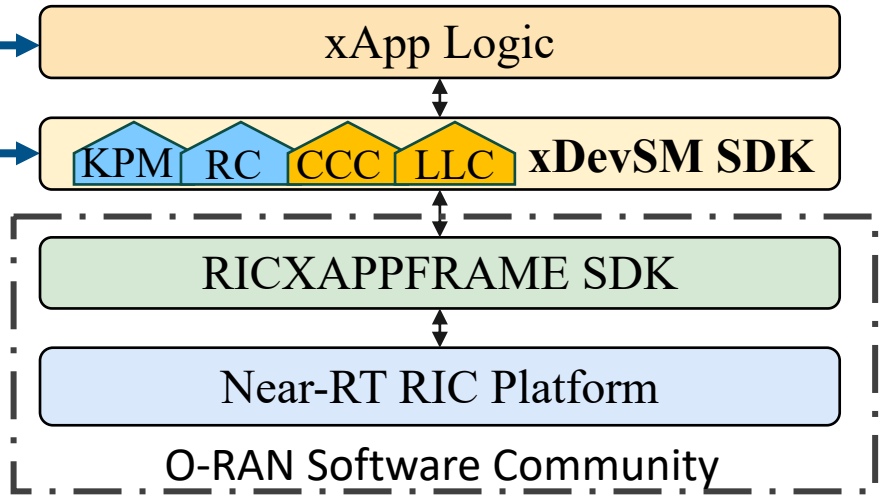
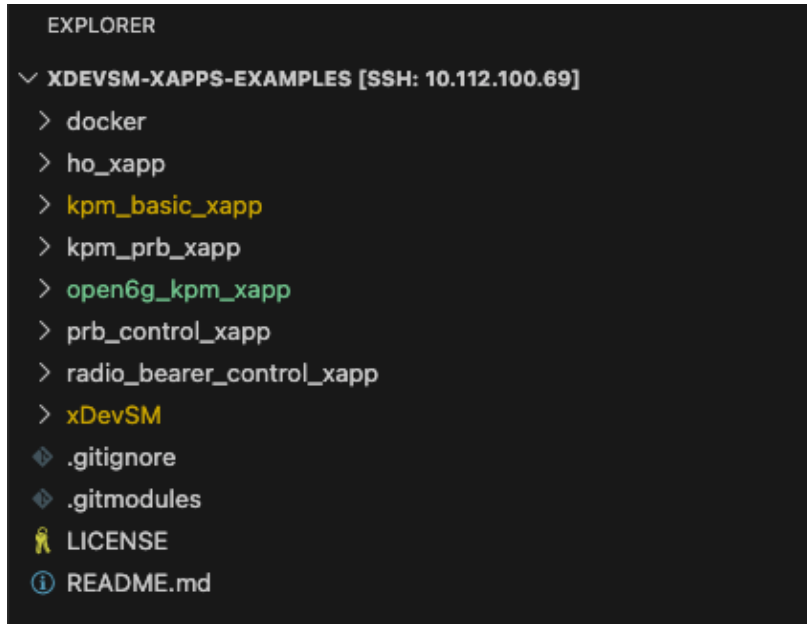
xApp Development Using xDevSM

xApps Repository

xDevSM-xapps-examples
(main module)
xDevSM
(submodule)



Branch *open6g-school*



What should we have inside the xApp pod?

- Content of the xApp directory
  - Config files
  - xApp Script
- xDevSM

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Our First xApp: Initialization

```
def main(args):

    # Creating a generic xDevSM RMR xApp
    xapp_gen = xDevSMRMRXapp("0.0.0.0", route_file=args.route_file)


    # Adding kpm functionalities to the xapp
    kpm_xapp = XappKpmFrame(xapp_gen,
                 xapp_gen.logger,
                 xapp_gen.server,
                 xapp_gen.get_xapp_name(),
                 xapp_gen.rmr_port,
                 xapp_gen.http_port,xapp_gen.get_pltnamespace(),
                 xapp_gen.get_app_namespace())


    # Registering the outermost rmr handler
    xapp_gen.register_handler(kpm_xapp.handle)


    # Registering indication message callback (not used)
    # kpm_xapp.register_ind_msg_callback(handler=indication_callback)


    # Registering termination signal handlers
    signal.signal(signal.SIGINT, kpm_xapp.terminate)
    signal.signal(signal.SIGTERM, kpm_xapp.terminate)
```

xApp Development

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Our First xApp: Getting E2 Node Information

```
gnb, gnb_info = xapp_gen.get_selected_e2node_info(args.gnb_target)
if gnb is None:
    xapp_gen.logger.error("No gNB selected. Exiting...")
    kpm_xapp.terminate(signal.SIGTERM, None)
    return
```

xApp Development

```
inventory_name: "gnb_001_001_00000e70"
global_nb_id {
  plmn_id: "00F110"
  nb_id: "000000000000000000000111001110000"
}
connection_status: CONNECTED
```

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Our First xApp: Getting E2 Node Information

```
gnb, gnb_info = xapp_gen.get_selected_e2node_info(args.gnb_target)
if gnb is None:
    xapp_gen.logger.error("No gNB selected. Exiting...")
    kpm_xapp.terminate(signal.SIGTERM, None)
    return


# Getting RAN Functions decoded
ran_function_description = kpm_xapp.get_ran_function_description(
                                                    json_ran_info=gnb_info)

func_def_dict = ran_function_description.get_dict_of_values()

xapp_gen.logger.debug("Available functions: {}".format(func_def_dict))
```

Available functions: {0: [], 1: [], 2: [], 3: ['DRB.PdcpSduVolumeDL', 'DRB.PdcpSduVolumeUL', 'DRB.RlcSduDelayDl', 'DRB.UEThpDl', 'DRB.UEThpUl', 'RRU.PrbTotDl', 'RRU.PrbTotUl'], 4: []}"

xApp Development

inventory_name: "gnb_001_001_00000e70"
global_nb_id {
  plmn_id: "00F110"
  nb_id: "000000000000000000000111001110000"
}
connection_status: CONNECTED

{
'ranName': 'gnb_001_001_00000e70',
'connectionStatus': 'CONNECTED',
'globalNbId': {
'plmnId': '00F110',
'nbId': '000000000000000000000111001110000'},
'nodeType': 'GNB',
'gnb': {
'ranFunctions': [{
'ranFunctionId': 2,
'ranFunctionDefinition': 60304F524...
...
}]
}

Our First xApp: Subscribing

```
# Only format 4 is supported by xDevSM
if len(func_def_dict[format_action_def_e.FORMAT_4_ACTION_DEFINITION]) == 0:
    xapp_gen.logger.error("Format 4 not supported. Exiting...")
    kpm_xapp.terminate(signal.SIGTERM, None)
    return

# Waiting for xApp registration (no callback available)
time.sleep(15)


# Sending subscription
ev_trigger_tuple = (0, args.event_trigger)
kpm_xapp.subscribe(gnb=gnb, ev_trigger=ev_trigger_tuple, func_def=func_def_dict,
ran_period_ms=1000, sst=args.sst, sd=args.sd)


# Start running after finishing subscription requests
xapp_gen.logger.info("Starting xapp")
xapp_gen.run()
```

xApp Development

Our First xApp: Config Files

```
{
"name": "open6g-kpm-xapp",
"version": "0.1.0",
"APP_NAMESPACE": "ricxapp-j",
"containers": [
    {
        "name": "open6g-kpm-xapp",
        "image": {
            "registry": "docker.io",
            "name": "angeloferaudo/open6g-kpm-xapp",
            "tag": "0.1.0"
            }
    }
    ],
    …
}
```

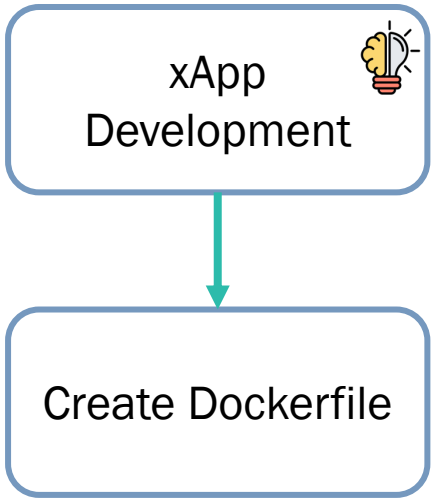
xApp Development

Our First xApp: Dockerfile

Key parameters that must be configured based on your specific setup:

- Working Directory
- **RIC Namespace**
- Entry Point

```
WORKDIR "/ws"
COPY open6g_kpm_xapp .
COPY xDevSM ./xDevSM
ENV PLT_NAMESPACE="ricplt-j"
# Automatic start
# CMD ["python3", "xapp.py"] # and args...
# Manual start
ENTRYPOINT ["sleep", "infinity"]
```

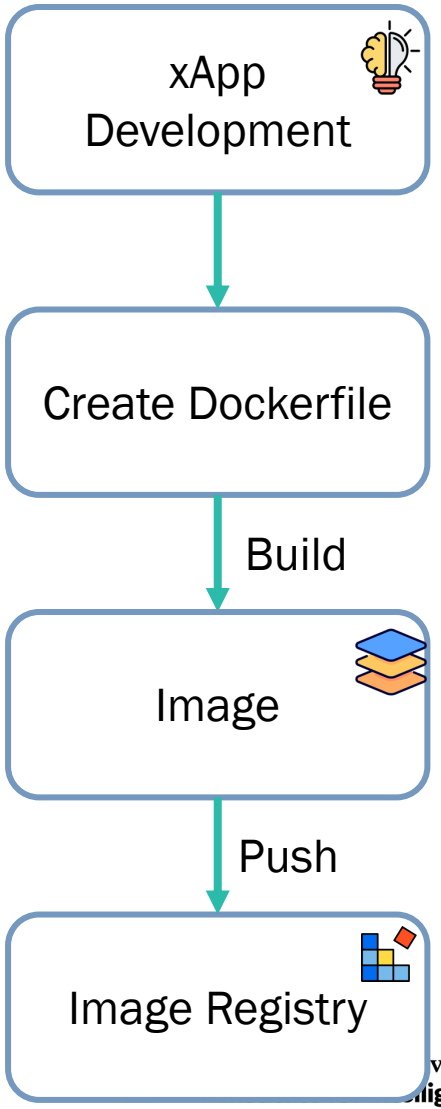


Our First xApp: Build and Push

This slide shows how to build and publish your xApp image to a public container registry (e.g., Docker Hub).

- Make sure to authenticate first using docker login

```
docker build --tag open6g-kpm-xapp:0.1.0 -file docker/Dockerfile.open6g_kpm_xapp .
docker tag open6g-kpm-xapp:0.1.0 angeloferaudo/open6g-kpm-xapp:0.1.0
docker push angeloferaudo/open6g-kpm-xapp:0.1.0
```



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xApp Deployment in OSC Near-RT RIC

OSC xApp Onboarding: dms_cli tool 1/2

The DMS (Deployment management services) tool (`dms_cli`) provides interfaces to onboard and deploy xApps on the OSC Near-RT RIC.

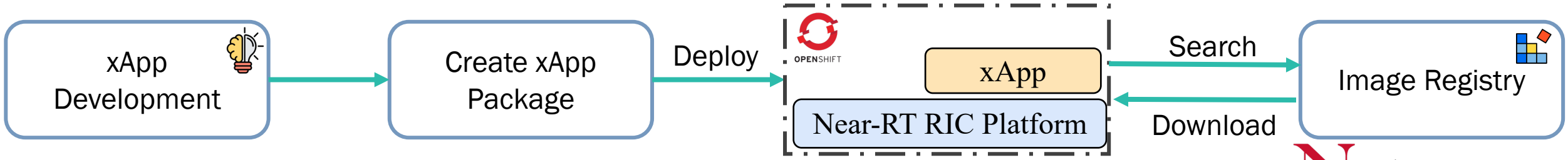
- Clone the repository https://github.com/o-ran-sc/ric-plt-appmgr.git
  - Checkout on the branch depending on the ric deployment
- To install the tool and its dependencies follow this guide
  - Activate the virtual environment after installation
- Once installed onboard the xApp

OSC xApp Onboarding: dms_cli tool 2/2

```
# Onboarding the xApp in chartmuseum
dms_cli onboard --config-file-path ~/xDevSM-xapps-examples/open6g_kpm_xapp/config/config-file.json --
shcema_file_path ~/xDevSM-xapps-examples/open6g_kpm_xapp/config/schema.json

# Download helm chart
dms_cli download_helm_chart open6g-kpm-xapp 0.1.0
# Result should be
# open6g-kpm-xapp-0.1.0.tgz

# Deploy the xApp
helm install open6g-kpm-xapp open6g-kpm-xapp-0.1.0.tgz -n ricxapp-j
```



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DEMO Closed-Loop Control using xDevSM 1

KPM xApp

RC xApp

DEMO Closed-Loop Control using xDevSM 2

root@demo-daniele-547f99597d-8rgmg: /mnt/homefolder/OAI_scripts (oc)

Initial gnb

```
UE 46fd: ulsch_rounds 1345/0/0/0, ulsch_errors 0, ulsch_DTX 0, BLER 0.00000 MCS (1) 0 (Qm 2 deltaMCS 0
 dB) NPRB 5  SNR 22.0 dB CCE fail 0
UE 46fd: MAC:    TX          17863 RX          43537 bytes
UE 46fd: LCID 1: TX           1228 RX           7479 bytes
UE 46fd: LCID 2: TX            100 RX            452 bytes
UE 46fd: LCID 4: TX           5064 RX           7424 bytes

[NR_RRC] Measurement Report for the serving cell 0 with RSRP: -86
[NR_RRC] Measurement Report for the serving cell 0 with RSRP: -86
[NR_RRC] Measurement Report for the serving cell 0 with RSRP: -86
[NR_MAC] Frame.Slot 640.0
UE RNTI 46fd CU-UE-ID 1 in-sync PH 48 dB PCMAX 21 dBm
UE 46fd: CQI 9, RI 2, PMI (11,1)
UE 46fd: dlsch_rounds 340/2/2/2, dlsch_errors 2, pucch0_DTX 8, BLER 0.00000 MCS (1) 0 CCE fail 9
UE 46fd: ulsch_rounds 1366/0/0/0, ulsch_errors 0, ulsch_DTX 0, BLER 0.00000 MCS (1) 0 (Qm 2 deltaMCS 0
 dB) NPRB 5  SNR 15.0 dB CCE fail 0
UE 46fd: MAC:    TX          18142 RX          44174 bytes
UE 46fd: LCID 1: TX           1234 RX           7563 bytes
UE 46fd: LCID 2: TX            100 RX            452 bytes
UE 46fd: LCID 4: TX           5157 RX           7522 bytes

[NR_RRC] Measurement Report for the serving cell 0 with RSRP: -86
[NR_RRC] Measurement Report for the serving cell 0 with RSRP: -86
[NR_RRC] Measurement Report for the serving cell 0 with RSRP: -86
```

dpugliese@dpugliese: ~ (ssh)

HO xApp

```
root@ho-control-xapp:/ws#
```

root@demo-daniele-1-869f59955d-p7sqx: /mnt/homefolder/OAI_scripts (oc)

Target gnb

```
[E2-AGENT]: E2 SETUP RESPONSE rx
[E2-AGENT]: Transaction ID E2 SETUP-REQUEST 0 E2 SETUP-RESPONSE 0
[UTIL]   threadCreate() for vnf_nvipc_aerial: creating thread with affinity ffffffff, priority 50
[VNF] Sent NFAPI_VNF_CONFIG_REQ num_tlv:34
TYPE <CTRL-C> TO TERMINATE
2992560.288323 [W] 2608787648: nr_error_ind_cb: [VNF] Received NFAPI_NR_PHY_MSG_TYPE_ERROR_INDICATION
idx:1 phy_id:0
2992560.288335 [W] 2608787648: nr_error_ind_cb: [VNF] Previous message 0x82 resulted in an error on th
e PNF
2992560.288337 [W] 2608787648: nr_error_ind_cb: [VNF] Received error code 0x34 (AERIAL_ERROR_CODE_MSG_
LATE_SLOT_ERR)
[NFAPI_VNF]
==========================================================================
sfn slot doesn't match unpacked one! L2->L1 0.0  vs L1->L2 0.4
==========================================================================
[NR_MAC] Frame.Slot 128.0

[NR_MAC] Frame.Slot 256.0

[NR_MAC] Frame.Slot 384.0

[NR_MAC] Frame.Slot 512.0

[NR_MAC] Frame.Slot 640.0
```

wineslab@winesnuc2-NUC: ~ (ssh)

UE

```
64 bytes from 10.45.0.1: icmp_seq=33 ttl=64 time=21.3 ms
64 bytes from 10.45.0.1: icmp_seq=34 ttl=64 time=13.7 ms
64 bytes from 10.45.0.1: icmp_seq=35 ttl=64 time=16.8 ms
64 bytes from 10.45.0.1: icmp_seq=36 ttl=64 time=18.9 ms
64 bytes from 10.45.0.1: icmp_seq=37 ttl=64 time=18.7 ms
64 bytes from 10.45.0.1: icmp_seq=38 ttl=64 time=19.7 ms
64 bytes from 10.45.0.1: icmp_seq=39 ttl=64 time=17.0 ms
64 bytes from 10.45.0.1: icmp_seq=40 ttl=64 time=15.5 ms
64 bytes from 10.45.0.1: icmp_seq=41 ttl=64 time=18.8 ms
64 bytes from 10.45.0.1: icmp_seq=42 ttl=64 time=18.4 ms
64 bytes from 10.45.0.1: icmp_seq=43 ttl=64 time=25.9 ms
64 bytes from 10.45.0.1: icmp_seq=44 ttl=64 time=24.0 ms
64 bytes from 10.45.0.1: icmp_seq=45 ttl=64 time=26.9 ms
64 bytes from 10.45.0.1: icmp_seq=46 ttl=64 time=24.9 ms
64 bytes from 10.45.0.1: icmp_seq=47 ttl=64 time=32.9 ms
64 bytes from 10.45.0.1: icmp_seq=48 ttl=64 time=30.9 ms
64 bytes from 10.45.0.1: icmp_seq=49 ttl=64 time=29.4 ms
64 bytes from 10.45.0.1: icmp_seq=50 ttl=64 time=27.0 ms
64 bytes from 10.45.0.1: icmp_seq=51 ttl=64 time=24.9 ms
64 bytes from 10.45.0.1: icmp_seq=52 ttl=64 time=24.2 ms
64 bytes from 10.45.0.1: icmp_seq=53 ttl=64 time=22.0 ms
64 bytes from 10.45.0.1: icmp_seq=54 ttl=64 time=19.9 ms
64 bytes from 10.45.0.1: icmp_seq=55 ttl=64 time=17.9 ms
64 bytes from 10.45.0.1: icmp_seq=56 ttl=64 time=20.9 ms
```

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References

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- O-RAN Specifications: https://specifications.o-ran.org/specifications
- Complete guide for this setup: https://openrangym.com/tutorials/xdevsm-tutorial
- xApp Deployment: https://github.com/aferaudo/ORANInABox/wiki
- xDevSM: https://github.com/wineslab/xDevSM
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